कर्नल राकेश खटकर
कमान अधिकारी
Col Rakesh Khatkar
Commanding Officer
Tele : 2508 (Off)
: 2539 (Resi)

19 अभियन्ता सैन्य दल
पिन : 914019
द्वारा 56 ए पी ओ
19 Engineer Regiment
PIN - 914019
c/o 56 APO

CORPS OF ENGINEERS

THE MADRAS SAPPERS

XIX

DARING TO TRAIN
TRAINING TO DARE

CHIEF OF THE ARMY STAFF
UNIT APPRECIATION : 2026

GENERAL OFFICER COMMANDING-IN-CHIEF WESTERN COMMAND
UNIT CITATION : 2021

उत्तराखण्ड शासन
CM APPRECIATION
OP RAHAT
2013

1241/DO/ 284 /A

20 March 2026

TO WHOM SO EVER IT MAY CONCERN

Dear Vineet Sharma

1. I am delighted to write this letter in appreciation of Mr Vineet Sharma and a strong endorsement of his abilities. I have actively monitored his work during his involvement in the project – "**Unmanned Ground Vehicle for Anti-Tank Mine Deployment**" carried out in collaboration with the 19 Engineer Regiment of the Indian Army.

2. Through this project, Vineet demonstrated leadership, technical depth, ownership and the capacity to lead, plan and execute a highly complex engineering task with multiple stakeholders.

3. Vineet played a leading role in the design and development of the semi-autonomous unmanned ground vehicle (UGV), developed to support mine laying operations in high altitude regions such as Ladakh, as well as other operational regions such as Rajasthan and Punjab.

4. The project entailed executing a multi-stage operation with three key objectives: ground excavation, mine placement and camouflage of the armed mine. This necessitated perfect coordination among the mechanical actuators, electronics control architecture, power system, and AI driven vehicle features.

5. Vineet led an 8-member team and played a pivotal role in planning, structuring, and driving the development efforts from an initial concept to the end product. The various phases included mechanical and system architecture design, fabrication and control systems testing, and field testing in alignment with operational requirements. What distinguished him was not only his technical competence, but also his ability to take ownership of critical functions that directly influenced the project's progress, coordination, and overall execution.

6. The initial phase included CAD design of the 4-wheel drive system, earth-auger-based digging mechanism, mine dropping mechanism, camouflage mechanism, control system architecture, and power and interface PCBs.

7. The fabrication of these mechanisms and control systems testing was carried out in the second phase. Control system architecture was established after accuracy, power analysis, and field needs.

8. The final phase included final fabrication, wiring, and control system integration. Initial field testing at the unit location examined runtime, power, accuracy, and stability. The UGV's electronic components were improved for field testing in sub-zero and high-altitude areas as per operational requirements.

9. Vineet was particularly involved in designing the entire control system architecture and carrying out subsequent embedded system development to support autonomous and semi-autonomous functionality within the UGV. This included designing the circuitry of the UGV, integrating various sensors, enabling communication, controlling BLDC hub motors and ESC, and implementing the power management system of the bot. He also designed interface and power PCBs in-house for application on the project.

10. In addition to technical and managerial aspects, he also handled fund and resource management for the entirety of the project. He analysed budget requirements, led procurement decisions, and ensured the effective use of available resources throughout the development cycle. His proactiveness during field testing enabled the team to make changes immediately as per on ground challenges and tailor the UGV as per operational requirements.

11. Vineet demonstrated initiative at the intersection of leadership, technical development, and resource management, while maintaining a solution-oriented approach. This was complemented by an exceptional work ethic to ensure completion of critical project milestones on time and as per required standard.

12. In my considered opinion, Vineet Sharma possesses the technical competence, intellectual maturity, leadership potential, and work ethic required to succeed in rigorous academic, research and professional settings. His performance on this project demonstrated a rare combination of analytical ability, execution discipline and ownership of responsibility enabling him to distinguish himself in any future opportunity he pursues.

13. ***The UGV based Anti Tank Mine Laying System developed by Vineet Sharma was selected among the top 30 innovations out of over 260 entries at the Northern Command Inno Yodha Competition 2025-26, underscoring the system's uniqueness and operational significance.***

Good Luck & God Speed!

With warm regards

Yours sincerely.